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Enhanced security for Banking Transactions using Image Based Steganography

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ARTICLE INFO ABSTRACT

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Abstract
In an age of increasing digital communication and data transfer, ensuring the security and privacy of sensitive information is paramount. Steganography, the art of hiding information within other data, has been used for centuries. In the digital realm, it plays a critical role in secure communication and information concealment. Traditional steganography methods often involve embedding information within a single image. While effective, this approach may be susceptible to detection, as single-image steganography can leave detectable traces, especially under sophisticated analysis. The primary challenge is to develop a robust system for multiple image steganography that can securely hide sensitive files within a set of images. This involves designing algorithms that distribute the information effectively across the images while maintaining imperceptibility and ensuring reliable extraction. Therefore, the rise of cyber threats and privacy concerns, there's a growing need for advanced techniques to protect sensitive files from unauthorized access or interception. Multiple image steganography, an emerging field, offers the potential for heightened security by spreading information across multiple images, making it even more challenging for potential adversaries to detect or extract. The project seeks to enhance file security by leveraging advanced techniques in multiple image steganography. By distributing the information across a set of images, this research endeavors to develop a system capable of securely concealing sensitive files. The algorithms utilized in this approach are designed to ensure imperceptibility and robustness against detection efforts. This advancement holds great promise for significantly improving the security of file transmission and storage, safeguarding critical information from unauthorized access or interception.
Keywords: data security, file concealment, multiple image steganography, privacy protection, secure communication, steganography algorithms, unauthorized access prevention

1. Introduction

In today's digital era, safeguarding sensitive information during communication and data transfer is of utmost importance. Throughout history, the practice of steganography, which involves concealing information within other data, has been utilized to achieve this goal. As we transition to digital mediums, steganography plays a crucial role in ensuring secure communication and information concealment. Traditionally, steganography focused on embedding information within a single image. While effective, this approach has its drawbacks, as it can be susceptible to detection, especially under sophisticated analysis. The challenge at hand is to develop a robust system for multiple image steganography that can securely hide sensitive files within a collection of images. This entails designing algorithms that efficiently distribute the information across the images while maintaining imperceptibility and ensuring reliable extraction.

The increasing prevalence of cyber threats and privacy concerns underscores the need for advanced techniques to protect sensitive files from unauthorized access or interception. Multiple image steganography, an emerging field, offers the potential for heightened security by dispersing information across multiple images. This makes it even more challenging for potential adversaries to detect or extract the concealed data.

The project aims to enhance file security by leveraging advanced techniques in multiple image steganography. The goal is to develop a system capable of securely concealing sensitive files by distributing information across a set of images. The algorithms employed in this approach are specifically designed to ensure imperceptibility and robustness against detection efforts. This advancement holds great promise for significantly improving the security of file transmission and storage, safeguarding critical information from unauthorized access or interception.

2. Literature Survey

B. Sultan,et.al[1] In this project The success of deep learning based steganography has shifted focus of researchers from traditional steganography approaches to deep learning based steganography. Various deep steganographic models have been developed for improved security, capacity and invisibility. In this work a multi-data deep learning steganography model has been developed using a well known deep learning model called Generative Adversarial Networks (GAN) more specifically using deep convolutional Generative Adversarial Networks (DCGAN). X. Liao,et.al[2] In this project With the coming era of cloud technology, cloud storage is an emerging technology to store massive digital images, which provides steganography a new fashion to embed secret information into massive images. Specifically, a resourceful steganographer could embed a set of secret information into multiple images adaptively, and share these images in cloud storage with the receiver, instead of traditional single image steganography.

M. Srivastava,et.al[3] In this project Image Steganography is the artwork of concealing mystery data within the image such that the hacker will now no longer be capable of discover the records within inside the stego images. This is a useful approach to secure our sensitive information. Security has continually been a main difficulty from last many years to existing days. The topic of interest to researchers has long been the development of secure technologies for sending data to anyone other than the recipient without revealing it. G. Benedict,et.al[4] In this project Steganography is the process of hiding a secret message within an ordinary message & extracting it at its destination. Image steganography is one of the most common and secure forms of steganography available today.

Traditional steganography techniques use a single cover image to embed the secret data which has few security shortcomings.

S. Mukhopadhyay,et.al[5] In this The paper proposes a scheme for achieving steganography with multiple encrypted monochromatic images with keys obtained from a synchronized system of semiconductor lasers. The key selection scheme for steganography determines the robustness of the application. It is in this area that steganography may benefit from the properties of chaos synchronization. A. S. Ansari,et.al[6] This paper presents an image Steganography algorithm that can work for cover images of multiple formats. Having a single algorithm for multiple image types provides several advantages. For example, we can apply uniform security policies across all image formats, we can adaptively select the most suitable cover image based on data length, network bandwidth and allowable distortions, etc.

P. Grandhe,et.al[7] In this project Communicating online without fearing third-party interventions is becoming a challenge in the modern world. Especially the sectors like the military, and government organizations or private companies sharing sensitive information. They invest a lot of effort and cost into obtaining the advancement of safe communication techniques. R. Joshi,et.al[8] in this There are advances in data stealth and forgery with the rise of technology and advances in data transmission. This arises the need for better and developed methods for data transmission. Data Transmission is an essential task in the current era, and equally important is the secure and safe information of that data. In the paper, batch steganography is used to secure data transmission from one end to the other. Often a password can be used for encoding the payload into the cover image. Here the data is encrypted using hashing and encryption techniques, SHA-256 and AES. The passwords used for encryption have been used after the logical operation XOR. Thus, the information has been encrypted twice using the XORed password for first and second input password for the next time.

Z. Wang,et.al[9] In this paper, a more accurate image steganography method is proposed, where a multi-level feature fusion procedure based on GAN is designed. Firstly, convolution and pooling operations are added to the network for feature extraction. Then, short links are used to fuse multi-level feature information. Finally, the stego image is generated by confrontation learning between discriminator and generator. Experimental results show that the proposed method has higher steganalysis security under the detection of high-dimensional feature steganalysis and neural network steganalysis. Comprehensive experiments show that the performance of the proposed method is better than ASDL-GAN and UT-GAN. X. Zhao,et.al[10]In this paper, a multi dilated generation countermeasure network (Multi Dilated GAN) model is proposed to improve the information steganography quality of images. Based on the discriminator of steganogan model, multiple convolution and expansion convolution are adopted. By selecting different expansion rates in the expansion convolution and matching with different receptive fields, different feature layers of the network can be effectively extracted. M. Liu,et.al[12] In this paper, They propose a new adversarial embedding scheme for image steganography. Unlike those related works, we first combine multiple gradients of cover and generated stego to determine the directions of cost modifications.

3. Proposed System

Figure 1 shows the image of steganography architecture.

Step 1: Steganography Dataset: The initial step in our research involves acquiring a suitable dataset for steganography. This dataset consists of various images that will serve as carriers for the hidden information. The selection of these images is crucial, as their characteristics can influence the effectiveness and imperceptibility of the steganographic technique. Typically, a diverse set of images with varying resolutions, color distributions, and complexities are chosen to ensure the robustness and generalizability of the proposed method.

Step 2: Dataset Preprocessing

Before utilizing the dataset, it is essential to preprocess the images to ensure they are suitable for steganographic embedding. This preprocessing includes:

- **Null Value Removal:** Ensuring that there are no corrupt or incomplete images in the dataset. Any image files with null values or missing data are either corrected or removed from the dataset.
- **Label Encoding:** For any categorical data associated with the images, such as image type or source, label encoding is performed to convert these categories into numerical values. This step is particularly useful if the dataset includes metadata that can influence the embedding process.

Step 3: Existing Least Significant Bit (LSB) Substitution

To establish a baseline for performance comparison, we first implement the traditional Least Significant Bit (LSB) substitution technique. In LSB steganography, the least significant bit of each pixel in an image is replaced with the bits of the secret message. This method is straightforward and widely used due to its simplicity and ease of implementation. However, it is also more susceptible to detection and less robust against image manipulations and attacks. By implementing this method, we can evaluate its strengths and weaknesses and set a benchmark for our proposed approach.

Step 4: Proposed Pixel Value Differencing (PVD) in Multiple Image Steganography

The core of our research lies in developing an advanced steganographic technique using Pixel Value Differencing (PVD) across multiple images. This involves several key steps:

- **Message Slicing:** The secret message is divided into smaller chunks, each of which will be embedded into different images. This distribution enhances security, as detecting and extracting the entire message requires access to all the carrier images.
- **PVD Embedding:** For each image, the PVD algorithm is applied. This method takes advantage of the differences in pixel values to embed information. By analyzing the differences between adjacent pixel values, the algorithm can embed bits of the secret message in a way that is less noticeable compared to altering individual pixel values directly.
- **Encoding and Storage:** The images with embedded data are saved, ensuring that the embedded information remains imperceptible to the naked eye and resilient against common image processing operations.

Step 5: Performance Comparison

To evaluate the effectiveness of our proposed PVD-based multi-image steganography technique, we perform a comprehensive performance comparison with the LSB substitution method. This comparison involves:

- **Imperceptibility:** Assessing the visual quality of the steganographic images to ensure that the embedded information is not noticeable. This is typically measured using metrics like Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM).

- **Robustness:** Testing the resilience of the embedded data against various attacks and manipulations, such as compression, resizing, and noise addition.
- **Capacity:** Comparing the amount of data that can be embedded without degrading the image quality.

Step 6: Prediction of Output from Test Data with Pixel Value Differencing (PVD) in Multiple Image Steganography Trained Model Algorithm

Finally, we develop and test a predictive model based on the PVD steganography technique. This involves:

- **Training the Model:** Using a portion of the preprocessed dataset, we train a model to embed and extract data using the PVD technique. The training process fine-tunes the algorithm parameters to optimize the embedding and extraction processes.
- **Testing and Validation:** The trained model is then tested on a separate set of images to validate its performance. We evaluate its accuracy in correctly embedding and extracting the secret message, as well as its robustness against various image alterations.
- **Analysis of Results:** The results from the test data are analyzed to assess the model's effectiveness. This includes evaluating the imperceptibility, robustness, and capacity of the steganographic method.

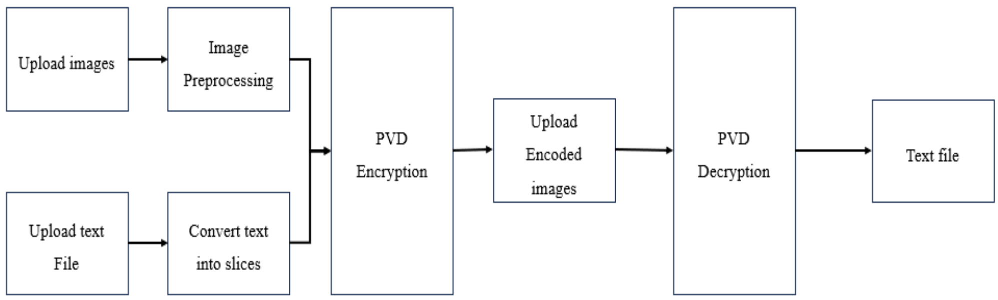

Figure 1: Architecture diagram of multiple image steganography

3.1 Pixel Value Differencing

Pixel Value Differencing (PVD) is an innovative approach to steganography that operates in the spatial domain of images. While PVD is commonly applied to single images, its extension to multiple image steganography introduces new dimensions of security and capacity. The fundamental concept of PVD revolves around manipulating the pixel values of multiple images to embed hidden information.

In PVD, the difference between the pixel values of adjacent pixels is utilized for data embedding. By carefully adjusting these differences, information can be hidden without significantly altering the visual appearance of the images. PVD operates in the spatial domain, making it resilient to frequency-based attacks. Unlike frequency domain techniques that might be susceptible to transforms, PVD directly modifies pixel values for data concealment.

The extension of PVD to multiple images involves distributing hidden information across a set of images. This approach enhances security by dispersing the embedded data, making it more challenging for adversaries to detect or extract the complete message.

Embedding Process:

The embedding process in PVD-based multiple image steganography follows a series of steps:

— Image Preparation: Select a set of cover images for embedding. These images act as carriers for different segments of the hidden message.

— Data Slicing: Divide the hidden message into segments corresponding to the number of selected cover images. Each segment is then embedded into the respective image.

— PVD Embedding: Apply the PVD algorithm to each image, adjusting pixel values based on the differences between adjacent pixels. The differences are manipulated to represent the hidden message.

— Secure Key Integration: Integrate a secure key into the embedding process to enhance security. The key determines how the pixel value differences are adjusted, and without it, extracting the hidden information becomes extremely challenging.

Extraction Process:

The extraction process is designed to retrieve the hidden message from the stego images:

— Image Selection: Choose the stego images containing segments of the hidden message.

— PVD Extraction: Apply the PVD algorithm in reverse to extract the differences between pixel values.

— Data Reconstruction: Reconstruct the hidden message segments from the extracted differences.

— Secure Key Utilization: Use the secure key during the extraction process to ensure accurate retrieval of the hidden information.

4. Results and Discussion

Figure 2 illustrates the input text that will be sliced into segments. Each segment is embedded into a different image from the uploaded folder. The sliced text ensures effective data distribution across multiple images. Figure 3 demonstrates the interface where users can upload a folder containing multiple images. These images serve as carriers for embedding the sliced text using PVD steganography.

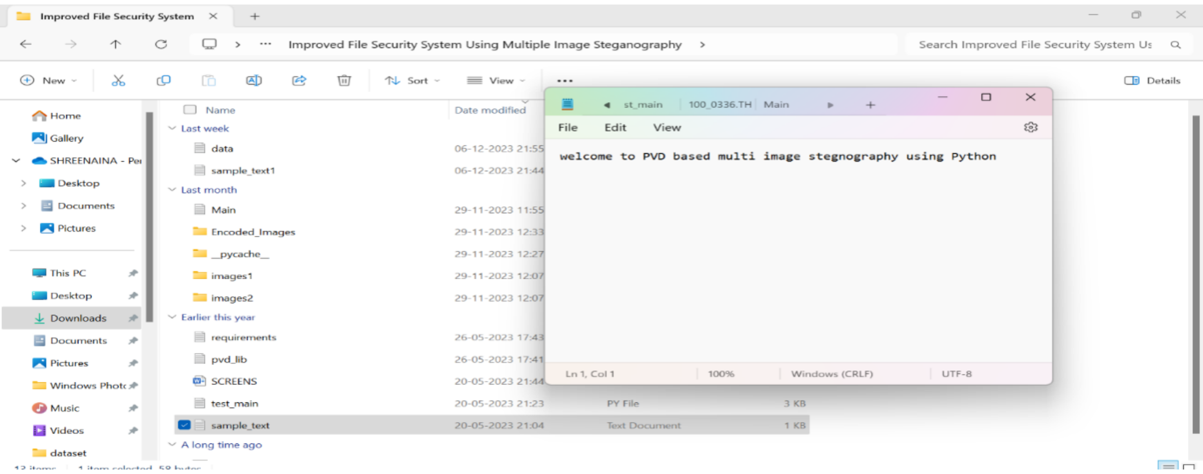

Figure 2. Above sample text will get sliced and hide inside multiple images

Figure 3. Upload Encoding Images

Figure 4 shows the step where a sample text file is uploaded for slicing and encoding. Each text slice is embedded into an image, and the encoded images are saved in the Encoded_Images folder. Figure 5 displays the slicing process, where each line of the text is associated with the name of the image in which it is hidden. For example, "welcome to PVD" is embedded in the first image, and subsequent slices are hidden in other images. Figure 6 illustrates the decoding process where a folder from Encoded_Images is uploaded. Upon selection, the embedded text slices are extracted and reconstructed into the original message.

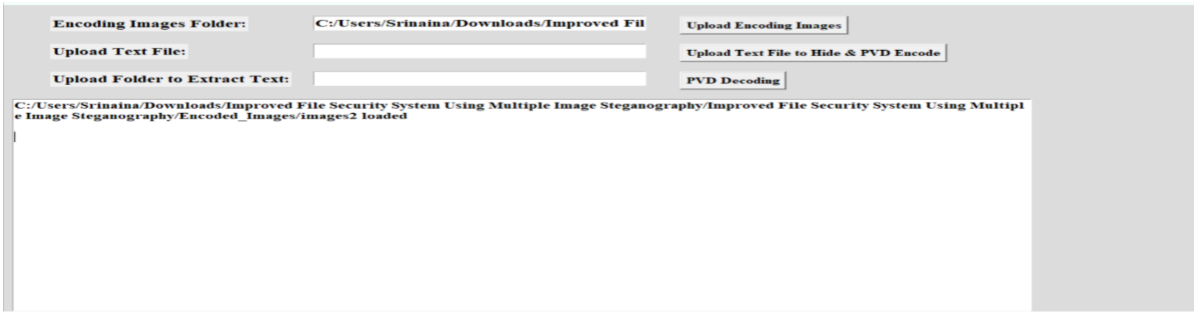

Figure 4. Upload Text File to Hide & PVD Encode.

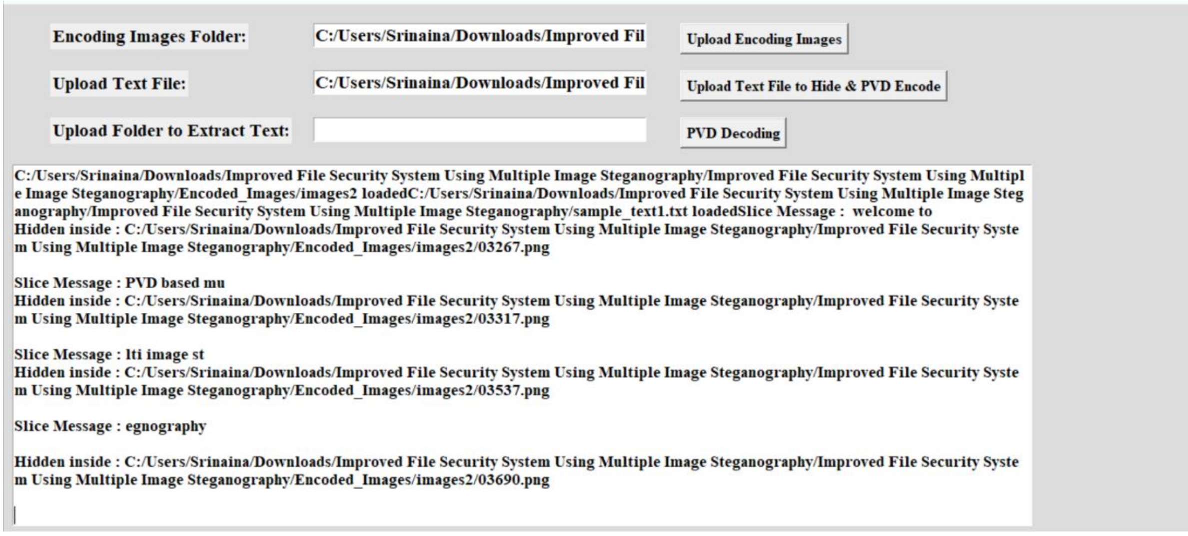

Figure 5. Slicing

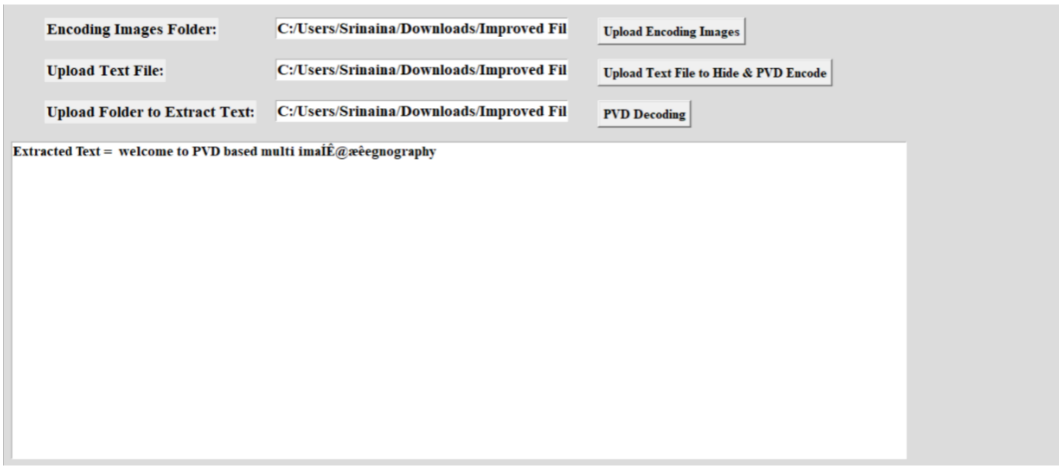

Figure 5. Extracted Message.

5. Conclusion

Multiple image steganography represents a significant advancement in the field of covert communication and secure data transmission. The technique of distributing hidden information across a series of images, coupled with the Pixel Value Differencing (PVD) algorithm, offers a potent combination of security, resilience, and imperceptibility. The strategic division of data, error-correction techniques, spread spectrum methods, and secret sharing schemes contribute to the robustness and reliability of the steganographic system. The incorporation of cryptography and encryption enhances the confidentiality of the concealed information, while authentication and watermarking techniques provide mechanisms for verifying the integrity of the images. Hybrid approaches, integrating various steganographic methods and security measures, offer adaptability and versatility to meet diverse security requirements. However, the implementation of multiple image steganography is not without challenges. Synchronization during the embedding and extraction processes is crucial for accurate data retrieval. Striking a balance between usability and security is a delicate consideration, requiring thoughtful trade-offs to create an effective and user-friendly steganographic system.

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